3-19-26 (WEEK 4)

Shun/翔海(@shun4midx)

EXAMPLE

Australian Holiday.

$\mathbb{P}(\text{sunny}) = 0.8$

$\mathbb{P}(\text{swim} \mid \text{sunny}) = 0.7$ $\quad \mathbb{P}(\text{swim} \mid \text{not sunny}) = 0.3$

$\mathbb{P}(\text{shark bite} \mid \text{swim}) = 0.5$ $\quad \mathbb{P}(\text{shark bite} \mid \text{not swim}) = 0.01$

$\mathbb{P}(\text{shark bite}) = ?$

By the Law of Total Probability, $\mathbb{P}(\text{shark bite}) = \mathbb{P}(\text{shark bite} \mid \text{swim})\mathbb{P}(\text{swim}) + \mathbb{P}(\text{shark bite} \mid \text{not swim})\mathbb{P}(\text{not swim})$

$= 0.5\,\mathbb{P}(\text{swim}) + 0.01\,[1 - \mathbb{P}(\text{swim})]$

$= 0.49\,\mathbb{P}(\text{swim}) + 0.01$

By the Law of Total Probability, $\mathbb{P}(\text{swim}) = \mathbb{P}(\text{swim} \mid \text{sunny})\mathbb{P}(\text{sunny}) + \mathbb{P}(\text{swim} \mid \text{not sunny})\mathbb{P}(\text{not sunny})$

$= 0.7(0.8) + 0.3(1 - 0.8)$

$= 0.62$

$\therefore\ \mathbb{P}(\text{shark bite}) = 0.49(0.62) + 0.01 = 0.3138$

Q: Given I get bitten by a shark, what is the probability it was sunny?

$$\mathbb{P}(\text{sunny} \mid \text{bite}) = \frac{\mathbb{P}(\text{sunny} \cap \text{bite})}{\mathbb{P}(\text{bite})} = \frac{\mathbb{P}(\text{bite} \mid \text{sunny})\,\mathbb{P}(\text{sunny})}{0.3138}$$

$\mathbb{P}(\text{bite} \mid \text{sunny}) = \mathbb{P}(\text{bite} \mid \text{swim} \cap \text{sunny})\mathbb{P}(\text{swim} \mid \text{sunny}) + \mathbb{P}(\text{bite} \mid \text{not swim} \cap \text{sunny})\,\mathbb{P}(\text{not swim} \mid \text{sunny})$

$= \mathbb{P}(\text{bite} \mid \text{swim})\,\mathbb{P}(\text{swim} \mid \text{sunny}) + \mathbb{P}(\text{bite} \mid \text{not swim})\,\mathbb{P}(\text{not swim} \mid \text{sunny})$

$= 0.5(0.7) + 0.01(1 - 0.7)$

$= 0.353$

$\therefore\ \mathbb{P}(\text{sunny} \mid \text{bite}) = \frac{0.353(0.7)}{0.3138} > 90\%$ 💀 :skull:

EXAMPLE (MONTY HALL PROBLEM)

Should the contestant switch?

<u>Events</u>

$F_1 = \{\text{initial choice is the car}\}$

$F_2 = \{\text{initial choice is a goat}\}$

$\mathbb{P}(F_1) = \frac{1}{3}$, $\mathbb{P}(F_2) = \frac{2}{3}$

$E = \{\text{wins if switch}\}$, $E' = \{\text{wins if no switch}\}$

By the Law of Total Probability, $\mathbb{P}(E') = \mathbb{P}(E' \mid F_1)\mathbb{P}(F_1) + \mathbb{P}(E' \mid F_2)\mathbb{P}(F_2) = 1 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \frac{1}{3}$

$\mathbb{P}(E) = \mathbb{P}(E \mid F_1)\,\mathbb{P}(F_1) + \mathbb{P}(E \mid F_2)\,\mathbb{P}(F_2) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}$

$\therefore$ Switch!

EXAMPLE (GAMBLER'S RUIN)

所以埋話賭錢賭得多會變碌柴咁撚樣

A gambler goes to a casino with \$67.

They need \$100 to pay debts.

They play a game where you win \$1 w/ prob $\frac{1}{2}$ and lose \$1 w/ prob $\frac{1}{2}$ (indep games)

As soon as they have \$100, they quit and are happy.

But if they reach \$0, they have to stop, and die.

$\mathbb{P}(\text{happy}) = ?$ ("wDyM tH:S :S a W:N wIn GaMe LOL")

<u>Solution</u>

Given $0 \le n \le 100$, let $p_n := \mathbb{P}(\text{happy} \mid \text{start with } \$n)$

We know $p_{100}=1$, $p_0=0$, $p_{67}=?$

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Consider the first round, then apply law of total probability

$\mathbb{P}(\text{happy} \mid \text{start with \$67}) = \mathbb{P}(\text{happy} \mid \text{win } 1^{st} \text{ round} + \text{start with \$67})\,\mathbb{P}(\text{win } 1^{st} \text{ round} \mid \text{start with \$67}) +$
$\mathbb{P}(\text{happy} \mid \text{lose } 1^{st} \text{ round} + \text{start with \$67})\,\mathbb{P}(\text{lose } 1^{st} \text{ round} \mid \text{start with \$67})$

By the independence of the rounds, $\mathbb{P}(\text{happiness})$ depends on how much money we have, NOT HOW we got there.

$= \mathbb{P}(\text{happy} \mid \text{start with \$68}) \cdot \frac{1}{2} + \mathbb{P}(\text{happy} \mid \text{start with \$66}) \cdot \frac{1}{2}$
$= \frac{1}{2}p_{68} + \frac{1}{2}p_{66}$

By the same reasoning, $\forall 1 \le n \le 99$, $p_n = \frac{1}{2}p_{n+1} + \frac{1}{2}p_{n-1}$ (recurrence relation)

Notice, $p_1 = \frac{1}{2}p_2 + \frac{1}{2}p_0 = \frac{1}{2}p_2$
$p_2 = \frac{1}{2}p_3 + \frac{1}{2}p_1 = \frac{1}{2}p_3 + \frac{1}{4}p_2 = \frac{2}{3}p_3$
$\vdots$

By induction, $p_n = \frac{n}{n+1}p_{n+1} \Rightarrow p_n = \frac{n}{n'}p_{n'}$, $n \le n' \le 100$

$\therefore p_n = \frac{n}{100}p_{100} = \frac{n}{100} \Rightarrow p_{67} = 0.67$

SECOND BOREL-CANTELLI LEMMA

RECALL

If $E_1, E_2, E_3, \ldots$ is a countably infinite sequence of events in $(S, \mathbb{P})$, then $\limsup_{n\to\infty} E_n = \bigcap_{n=1}^{\infty}\left(\bigcup_{i=n}^{\infty} E_i\right)$

FIRST BCL

$\sum_{n=1}^{\infty}\mathbb{P}(E_n) < \infty \Rightarrow \mathbb{P}(\limsup_{n\to\infty} E_n) = 0$

THEOREM (SECOND BCL)

Let $(S, \mathbb{P})$ be a probability space and let $E_1, E_2, E_3, \ldots$ be a countably infinite sequence of independent events. Then, $\sum_{n=1}^{\infty}\mathbb{P}(E_n) = \infty \Rightarrow \mathbb{P}(\limsup_{n\to\infty} E_n) = 1$

<u>Proof</u>

By def, $\mathbb{P}(\limsup_{n\to\infty} E_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty}\left(\bigcup_{i=n}^{\infty} E_i\right)\right) = \lim_{n\to\infty}\mathbb{P}\left(\bigcup_{i=n}^{\infty} E_i\right)$

$= 1 - \lim_{n\to\infty}\mathbb{P}\left(\bigcap_{i=n}^{\infty} E_i^c\right)$ (complement + DeM)

$= 1 - \lim_{n\to\infty}\prod_{i=n}^{\infty}\mathbb{P}(E_i^c)$ (indep)

$= 1 - \lim_{n\to\infty}\prod_{i=n}^{\infty}(1 - \mathbb{P}(E_i))$ (complement)

Recall the convergence test for inf products: $\sum_{i=n}^{\infty}\mathbb{P}(E_i) = \infty \Rightarrow \prod_{i=n}^{\infty}(1 - \mathbb{P}(E_i)) = 0$

$\therefore \mathbb{P}(\limsup_{n\to\infty} E_n) = 1 - 0 = 1$ □

DISCRETE RANDOM VARIABLES

DEFINITION

Given a probability space $(S, \mathbb{P})$, a random variable is a function $X: S \to \mathbb{R}$

We say it is discrete if it only takes countably many values

The random variable determines a (usually simpler) probability space on the set of random values it takes, $X(S)$

$\forall x \in S$, we say $\{X = x\} \subseteq S$
$\{X = x\} = \{w \in S \mid X(w) = x\}$

If X is discrete, let $x_1, x_2, x_3, \ldots$ be the values it takes.

Then, $S = \bigcup_{i=1}^{\infty}\{X = x_i\}$, and we can define a distribution $p(x_i) = \mathbb{P}(X = x_i) = \mathbb{P}(\{w \in S \mid X(w) = x_i\})$

This defines a new probability space $(\{x_1, x_2, \ldots\}, p)$, where p is called the probability mass function of X.

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EXAMPLE

Multiple choice test:

- 5 Qs, each with 4 options, one is correct
- You choose a uniform random answer for each question independently
- $X = \#$Qs correct

What is $p(k)$ $\forall 0 \le k \le 5$?

<u>Solution</u>

Since our choices are uniformly random,

$$p(k) = \mathbb{P}(\{k \text{ answers correct}\}) = \frac{\#\{\text{choices w/ } k \text{ correct}\}}{\#\{\text{choices}\}} = \frac{|\{\text{choices w/ } k \text{ correct}\}|}{4^5}$$

To count sequences w/ exactly k correct,

① Choose which k answers will be correct: $\binom{5}{k}$ options

② Choose the wrong answers: 3 options for each

$\therefore$ $\#\{\text{choose exactly } k \text{ correct}\} = \binom{5}{k}3^{5-k}$

$$\Rightarrow p(k) = \frac{\binom{5}{k}3^{5-k}}{4^5} = \binom{5}{k}\left(\frac{1}{4}\right)^k\left(\frac{3}{4}\right)^{5-k}$$

Alternatively, $\mathbb{P}(\text{getting exactly } k \text{ right}) = \mathbb{P}\left(\bigsqcup_{R \subseteq [5], |R|=k} \{\text{ans in } R \Leftrightarrow \text{ans correct}\}\right)$

$$= \sum_{R \subseteq [5], |R|=k} \mathbb{P}(\{\text{ans in } R \Leftrightarrow \text{ans correct}\})$$

(Let $E_i = \{\text{ans } i \text{ correct}\}$, $\mathbb{P}(E_i) = \frac{1}{4}$, $\mathbb{P}(E_i^c) = \frac{3}{4}$)

$$= \sum_{R \subseteq [5], |R|=k} \mathbb{P}\left(\left(\bigcap_{i \in R} E_i\right) \cap \left(\bigcap_{i \notin R} E_i^c\right)\right)$$

By indep, $= \sum_R \prod_{i \in R} \mathbb{P}(E_i) \cdot \prod_{i \notin R} \mathbb{P}(E_i^c)$

$$= \binom{5}{k}\left(\frac{1}{4}\right)^k\left(\frac{3}{4}\right)^{5-k}$$